

Data-Driven Social Engagement Initiative

Major Project / Capstone Project

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Course: Data Science

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PROBLEM STATEMENT

Content creators and digital marketers lack a systematic, data-driven approach to understanding why certain content performs better than others. Decisions around posting time, content type, and topic selection are largely made on intuition rather than evidence. This results in inconsistent engagement, wasted effort, and missed growth opportunities on social media platforms.

OBJECTIVE

To design and build a complete end-to-end Machine Learning analytics ecosystem that automatically collects social media performance data, applies statistical and ML models to extract actionable insights, and delivers strategy recommendations through a 9-page interactive dashboard — demonstrating the full data science lifecycle from ingestion to serving.

MODULES BUILT

- **Module 1 — Content Performance Tracker:** Automated data extraction via YouTube Data API v3. Fetches video metadata, engagement metrics, and audience comments. Stores structured data in CSV and SQLite database with a hybrid real + dummy data mode for development.
- **Module 2 — Viral Score Engine:** Computes a 0–100 Viral Score per video using a weighted min-max normalisation formula across five engagement dimensions: views, likes, comments, like rate, and comment rate.

- **Module 3 — NLP Sentiment Analyser:** TextBlob-based sentiment classification of audience comments into Positive, Neutral, and Negative categories with polarity scores (-1 to +1) and subjectivity scoring.
- **Module 4 — A/B Testing Framework:** Statistical hypothesis testing using Welch's t-test (unequal variance), Cohen's d effect size measurement, and Lift % calculation across four content variables: posting time, video length, tag count, and description length.
- **Module 5 — Virality Prediction Model:** XGBoost Regressor trained on 16 engineered features to predict viral score from video metadata. Achieved $R^2 = 0.84$ on test set and Cross-Validation $R^2 = 0.94 \pm 0.02$ across 5 folds.
- **Module 6 — Content Recommender:** TF-IDF vectorisation on top-quartile video titles extracts high-performing keywords. Generates 7 ranked, confidence-scored strategy recommendations with evidence-backed rationale.
- **Module 7 — Trend Forecaster:** Facebook Prophet model with Holt-Winters ETS fallback for 90-day engagement forecasting with 80% confidence intervals, plus keyword trend velocity analysis (Rising / Falling / Stable).

TECH STACK

Layer	Tools
Data	Python, Pandas, NumPy, SQLite
API	YouTube Data API v3, Requests
NLP	TextBlob, Scikit-learn TF-IDF
Machine Learning	XGBoost, Scikit-learn, SciPy
Forecasting	Facebook Prophet, Statsmodels ETS
Dashboard	Streamlit, Plotly, Matplotlib
Deployment	Docker, Streamlit Cloud
Utilities	Loguru, Joblib, Python-dotenv

KEY RESULTS

0.84 XGBoost R ²	0.94 Cross-Val R ²	9 Dashboard Pages
90	7	35

Day Forecast	Recommendations	Source Files
4,745 Lines of Code		

OUTCOME

A fully functional, production-deployed ML analytics platform demonstrating the complete data science lifecycle: data ingestion → feature engineering → model training → evaluation → serving. The system is packaged as a Docker container and an interactive Streamlit web application, and generates a self-contained 181KB HTML analytical report automatically.

Resume Description: Built an end-to-end ML analytics ecosystem integrating YouTube Data API v3, TextBlob NLP sentiment classification, XGBoost virality prediction ($R^2=0.84$), Welch's t-test A/B testing framework, TF-IDF content recommendation engine, and Facebook Prophet 90-day trend forecasting. Delivered a 9-page interactive Streamlit dashboard with real-time ML predictions. Deployed via Docker and Streamlit Cloud.